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Static Casimir effect induced optical chaos in an optomechanical system

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Abstract

Static Casimir effect as one of the most interesting purely quantum effects is very general and may bring about novel physical phenomena and find applications in various fields of physics. Here, we investigate theoretically the generation of optical chaos induced by the static Casimir effect in an optomechanical system. We show that the dynamic behavior of the cavity field experiences the conversion from period to chaos motion under the condition of properly adjusting the static Casimir effect, even if the driven optical fields are comparatively weak. Furthermore, our numerical calculation with experimentally attainable parameters also shows that compared with the standard bare optomechanical system, the optical threshold of chaos is reduced about one order when the static Casimir effect is applied. Beyond the significance in broadening the scope of cavity optomechanics, this finding provides a different perspective to the study of optical chaos and lays a foundation for using static Casimir effect to comprehend the nonlinear characteristics of the optomechanical system. In particular, our results may find potential applications in chaos-based communication.

Keywords: cavity optomechanics, optical chaos, static Casimir effect

(Some figures may appear in colour only in the online journal)

1. Introduction

Benefiting from the rapid progress of nanotechnology and the unparalleled capacity of manufacturing materials on the nanometer scale in recent years, the study of the Casimir effect has become one of the most advanced subjects in scientific research. As one of the most interesting purely quantum effects, Casimir effect has gained growing attention and investigation [1–6], and the corresponding Casimir force has been precisely measured by many groups [7–9]. It is well known that the Casimir effect mainly includes two important effects in quantum optics: static Casimir effect and dynamical Casimir effect. Studies of the dynamical Casimir effect have been carried out by domestic and foreign scholars [4, 10–16]. The significance of the static Casimir effect [17, 18] in nano-devices is underscored due to the space separation between the component surfaces is dramatically decreased. Parallel to this, cavity optomechanics is a rapidly growing subject, which explores the radiation-pressure-based optomechanical interaction between the mechanical vibration and the optical cavity field [19–37]. In the context of the optomechanical system, the static Casimir effect between the moveable mirror (mechanical resonator) and the sphere involves the change of the mechanical resonator position depending on the sphere-mirror separation. This provides an alternative platform for tailoring the coupling between the cavity field and the mechanical resonator, bringing about some interesting phenomena and effects. For example, nonlinear mechanical oscillations [38], optomechanical entanglement [39], and optomechanically induced transparency [40] have been investigated. Besides that, the novel effects induced by the Casimir effect may find practical applications in quantum sensing of motion [41].

Recently, due to the intrinsic nonlinearity of optomechanical interaction and its prominent application in optical comb, the study of nonlinear interactions has attracted enormous

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attention and enabled many intriguing topics, such as high-order sideband generation [42–44], sum- and different- sideband generation [45, 46], optical frequency combs [47, 48], and blockade [49, 50]. In particular, as a widely existing nonlinear dynamic phenomenon, chaos is a motion that obeys the deterministic law but has randomness, namely, chaos is an aperiodic motion in a deterministic system, which features a sensitive dependence on tiny perturbations [51]. Chaotic motion is the unity of opposites between determinateness and randomness, that is, it has randomness but is not real or complete random motion. However, the random motion does not obey any dynamic equation, so its evolution with time is completely unpredictable. Chaotic motion occurs in a deterministic system, which is essentially different from completely random [52]: (i) chaotic motion obeys certain dynamic laws; (ii) although chaotic oscillation is random and irregular, its motion is not completely random; (iii) although chaotic motion is random in the whole time process, namely, it can not be predicted in a long time, but in a short time range, prediction is still possible. As we all know, the nonlinear system is frequently accompanied by chaotic motion as the nonlinear strength is strong enough and reaches the threshold of chaotic [53–57]. Without loss of generality, chaotic dynamics of the cavity optomechanical system is usually expressed by the semiclassical Langevin equations of motion. To our knowledge, period-doubling bifurcation is one of the momentous ways to chaos, that is, the occurrence of period-doubling bifurcation portends the presence of chaos [58]. Exploration of optical chaos and the correlated nonlinear dynamics promotes the advance of science and technology. Realizing and controlling an optical chaos is of fundamental significance in nonlinear science [59, 60], and may find span-new applications in chaotic encryption [61] and chaos-based secure communication [62]. The generation of optical chaos induced by the static Casimir effect, however, is still in the initial gradation of optomechanical system. Accordingly, the study of optical chaos generation induced by the static Casimir effect is probably of great significance for broadening the scope of cavity optomechanics, enriching the theory of optical chaos in optomechanical system and promoting the development of optomechanical physics. Most importantly, this work provides a different perspective to the study of optical chaos and lays a foundation for using static Casimir effect to comprehend the nonlinear characteristics of the optomechanical system.

Motivated by these works mentioned above, in this work, optical chaos generation in an optomechanical system coupled to a nanosphere is systematically investigated. The results of our numerical calculation with experimentally attainable parameters clearly demonstrate that optical chaos generation can be obtained via properly adjusting the static Casimir effect, even if the driven optical fields are comparatively weak. More interestingly, our numerical calculation also shows that the time domain waveform of cavity field may present periodic motion, periodic-doubling motion and aperiodic motion for diverse static Casimir effect. Regarding the significant role of optical chaos, the presence of static Casimir effect may provide a novel and valuable method for the operation of light. We believe that optical chaos induced by static Casimir effect may imply great potential application on chaos-based communication.

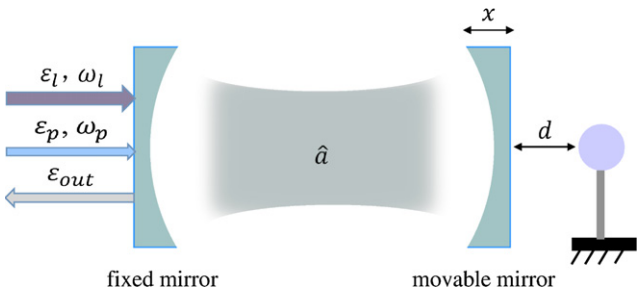


Figure 1. Schematic diagram of an optomechanical system with a tunable static Casimir effect. The cavity is linearly coupled to the displacement of a moveable mirror (mechanical resonator) via the radiation pressure. And the mechanical resonator interacts with the nearby gold-coated nanosphere via static Casimir effect. The optical cavity mode is simultaneously driven by a strong control field of amplitude ε_l with frequency ω_l and a weak probe field of amplitude ε_p with frequency ω_p.

This paper is structured as follows: in section 2, we present the theoretical model of the optomechanical system coupled to a nanosphere and give the semiclassical Heisenberg–Langevin equations of motion. In section 3, we analyze the characteristics of the cavity field dynamic behavior and discuss in detail the influence of static Casimir effect on optical chaos generation. Finally, a conclusion is summarized in section 4.

2. Physical model and dynamical equation

The physical model we consider is schematically shown in figure 1 where the mechanical resonator interacts with the nearby gold-coated nanosphere via the static Casimir effect. In the sphere-plane configuration, the zero-point Casimir force for a fixed sphere-plate space separation d can be written as [63] $F_c = \frac{2\pi^3\hbar cR}{720d^3}[1 + (2\beta - 1)\frac{d}{R}]$ with $\beta = \frac{2}{3} - \frac{10}{\pi^2}$. Here the first term represents the complete reflection expression in the proximity force approximation (PFA). The second term indicates the low-order modification to the PFA [64]. R and c are the radius of nanosphere and the speed of light in vacuum, respectively. In the case of $d \ll R$ (the PFA is valid), the zero-point Casimir force can be reduced to $F_c = 2\pi^3\hbar cR/(720d^3)$. Simultaneously, the system is driven by a relatively weak probe field with the frequencies ω_p and a strong control field with frequency ω_l. Thus, in a frame rotating at frequency ω_l, the Hamiltonian of our scheme can, therefore, be written as [40]

$$H = \hbar\Delta\hat{a}^\dagger\hat{a} + \left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega_m^2\hat{x}^2\right) - \frac{\pi^3\hbar cR}{720(d-\hat{x})^2} - \hbar g\hat{a}^\dagger\hat{a}\hat{x} + i\hbar\left[\left(\varepsilon_l + \varepsilon_p e^{-i\Omega t}\right)\hat{a}^\dagger - \text{H.c.}\right] \quad (1)$$

where the first term represents the free Hamiltonian of the cavity field with $\Delta = \omega_a - \omega_l$ and $\hat{a}^\dagger$ ($\hat{a}$) being the cavity-control field detuning and the creation (annihilation) operators of the cavity field. Generally, the creation and annihilation operators satisfy the commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$. The second

term denotes the energy of the mechanical resonator with $\hat{p}$ and $\hat{x}$ being the momentum and position operators of mechanical resonator. The third term describes the interaction between the mechanical resonator and the nanosphere, namely, Casimir energy, in which R, d, and c are the radius of nanosphere, the separation between the nanosphere and the mechanical resonator, and the speed of light in vacuum, respectively. It is worth noting that the space separation between the nanosphere and the mechanical resonator is significantly smaller than the dimension of adjacent plane [40]. The fourth term indicates the interaction between the cavity field and the mechanical resonator via the radiation pressure with g being the optomechanical coupling strength. The fifth term expresses the excitation of the external fields to the cavity mode with $\Omega = \omega_{\rm p} - \omega_{\rm l}$ being the probe-control field detuning. And the amplitudes of the driving fields defined as $\varepsilon_{\rm l} = \sqrt{2\kappa P_{\rm l}/\hbar\omega_{\rm l}}$ and $\varepsilon_{\rm p} = \sqrt{2\kappa P_{\rm p}/\hbar\omega_{\rm p}}$ with $P_{\rm l}$, $P_{\rm p}$, and κ being the power of the control field, the power of the probe field, and the total decay rate of the cavity, respectively.

In this work, we mainly focus on the mean response of the system, so the operators can be reduced to their expectation values, viz, $\alpha(t) \equiv \langle\hat{\alpha}(t)\rangle$ where α expresses any one of these quantities p, x,and a. To investigate the dynamic behavior of the optomechanical system with a view toward chaotic motion, we, therefore, derive the semiclassical Heisenberg–Langevin equations of motion as

$$\begin{aligned}
\dot{p} &= -m\omega_{\rm m}^2 x + \hbar g(a_{\rm r}^2 + a_{\rm i}^2) - \frac{2V_{\rm sp}}{(d-x)^3} - \gamma_{\rm m} p \\
\dot{x} &= \frac{p}{m}, \\
\dot{a}_{\rm r} &= (\Delta - gx)a_{\rm i} - \kappa a_{\rm r} + \varepsilon_{\rm l} + \varepsilon_{\rm p}\cos(\Omega t), \\
\dot{a}_{\rm i} &= (-\Delta + gx)a_{\rm r} - \kappa a_{\rm i} - \varepsilon_{\rm p}\sin(\Omega t),
\end{aligned} \tag{2}$$

Here $V_{\rm sp} = \frac{-\pi^3\hbar c R}{720}$, $\gamma_{\rm m}$ being the decay rate of the mechanical resonator, and the quantum noise terms can be dropped safely due to their expectation values being zero in the semiclassical approximation [21]. Moreover, the mean value of the cavity field operator is defined as $a = a_{\rm r} + {\rm i}a_{\rm i}$ ($a_{\rm r}$ and $a_{\rm i}$ are real numbers), which may facilitate the investigation of optical chaos. Distinctly, we can know from equation (2) that the mechanical displacement is affected by static Casimir effect. By further analyzing the evolution equations in equation (2), also we can find that the mechanical displacement is coupled from the intracavity field amplitude. In other words, the intracavity field amplitude and the mechanical displacement are mutually influenced in the process of evolution via the optomechanical coupling. Traditionally, this is responsible for the emergence of chaotic dynamics of the cavity optomechanical system at a comparative high-power laser-driving [54, 55, 66]. Nevertheless, unlike the previous study, the present optomechanical system is also affected by the static Casimir effect. It is common knowledge that the butterfly effect is a notable example of chaotic motion, shows that chaos system is highly sensitive to tiny disturbances under the initial conditions, and its trajectory in phase space is unrepeatable and unpredictable [67].

To explore the random features of our system, we assume that the evolution of a perturbation is $\vec{\delta} = (\delta x, \delta p, \delta a_{\rm r}, \delta a_{\rm i})$ which characterizes the divergence of adjacent trajectories in phase space. Thus, $\vec{\delta}$ can be obtained by linearizing equation (2) and can be written as

$$\dot{\vec{\delta}} = M\vec{\delta} \tag{3}$$

with coefficient matrix

$$M = \begin{pmatrix} 0 & \frac{1}{m} & 0 & 0 \\ -m\omega_{\rm m}^2 - \frac{6V_{\rm sp}}{(d-x)^4} & -\gamma_{\rm m} & 2\hbar g a_{\rm r} & 2\hbar g a_{\rm i} \\ -g a_{\rm i} & 0 & -\kappa & \Delta - gx \\ g a_{\rm r} & 0 & -\Delta + gx & -\kappa \end{pmatrix}, \tag{4}$$

in which $\delta a_{\rm r}$ and $\delta a_{\rm i}$ are the perturbations of the real and imaginary part of a, respectively. Significantly, $a_{\rm r}$, $a_{\rm i}$ and x which appears in the drift matrix M are time-dependent rather than steady-state values. We can use the Runge–Kutta method to solve equations (2) and (3), and the initial conditions are $\vec{o}_0 = (x, p, a_{\rm r}, a_{\rm i}) = (0, 0, 0, 0)$. Given this, we introduce the intensity of the cavity field and define it as $I_{\rm a} = a_{\rm r}^2 + a_{\rm i}^2$, its power spectrum $S(\omega)$ can be obtained via making fast Fourier transform of $I_{\rm a}$, where ω represents the spectroscopy frequency [44, 68].

3. Results and discussion

In this section, we first demonstrate the generation of periodic and chaotic motion in an optomechanical system with a tunable static Casimir effect. Also, we study the connection between the periodic and chaotic motion. Then we investigate the influence of static Casimir effect on the optical threshold of chaos.

To understand the dynamic characteristics of the optomechanical system with a tunable static Casimir effect, we present the oscillation of the cavity field in the frequency domain and temporal domain under different static Casimir effect in figure 2. It is shown that from figure 2(a), the generation of high-order sidebands with equal spacing in the frequency domain reveals the nonlinear characteristics of the coupled system. It is important to note that the static Casimir effect does not exist here, i.e., $d = \infty$. As we expected, the sideband spacing between the adjacent sideband lines is consistent with the beat frequency between the control field and the probe field. In this case, the oscillation of the cavity field in the temporal domain is stationary as is shown in figure 2(d), which means that the evolution of the cavity field is regular. In figure 2(b), we add the static Casimir effect with $d = 3.5$ nm which shows that the sideband spacing can be adjusted by regulating the sphere-mirror separation d. Furthermore, it is also shown that the period of the cavity field oscillation is extended relative to the figure 2(a), which is the so-called period-doubling bifurcation oscillation. And such period-doubling bifurcation oscillation is obviously observed in figure 2(e). To further investigate the influence of the static Casimir effect on the dynamic features of the cavity field, we decrease the sphere-mirror separation

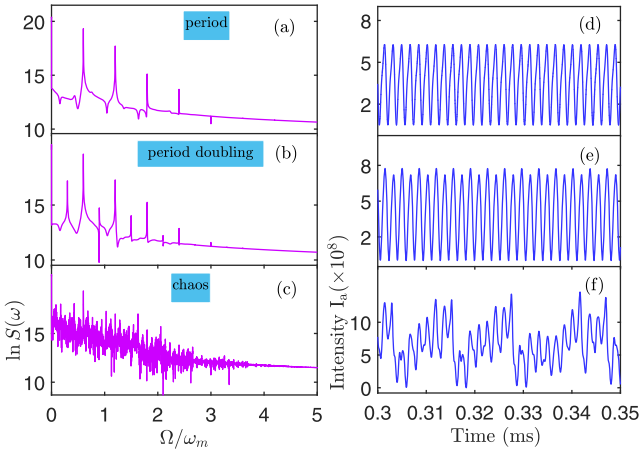


Figure 2. Oscillation of the optical cavity field is plotted in the frequency domain (a)–(c) as well as in the temporal domain (d)–(f), for three different values of the sphere-mirror separation $d = \infty$, $d = 3.5$ nm, $d = 2.5$ nm, respectively. The other specific parameters are adopted from the experiment in references [23, 65] as $m = 145$ ng, $\omega_{\rm m} = 2\pi \times 947$ KHz, $\gamma_{\rm m} = 2\pi \times 141$ KHz, $\kappa = 2\pi \times 215$ KHz, $R = 150$ nm, $g/2\pi = 70$ MHz nm^{-1}, $\lambda_{\rm l} = 2\pi c/\omega_{\rm l} = 1064$ nm, $P_{\rm l} = 149.5$ μW, $P_{\rm p} = 37.4$ μW, $\Omega = 0.6\omega_{\rm m}$ and $\Delta = \omega_{\rm m}$. The initial value is $\vec{o}_0 = (x, p, a_{\rm r}, a_{\rm i}) = (0, 0, 0, 0)$, and the perturbation is $\vec{\delta} = (\delta x, \delta p, \delta a_{\rm r}, \delta a_{\rm i}) = (10^{-20}, 10^{-20}, 10^{-13}, 10^{-13})$.

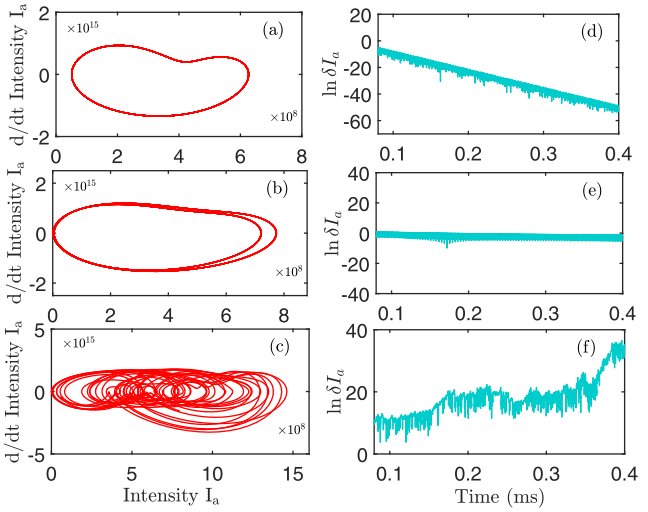


Figure 3. Evolution of the optical cavity field in phase space (a)–(c) and $\ln \delta I_{\rm a}$ in temporal domain (d)–(f) for three different values of the sphere-mirror separation $d = \infty$, $d = 3.5$ nm, $d = 2.5$ nm, respectively. The other parameters are the same as those in figure 2.

d to 2.5 nm, and the oscillation of the cavity field in the frequency domain and temporal domain are shown in figures 2(c) and (f), respectively. We uncover from figures 2(c) and (f) that the equal spacing sideband spectrum converts to a continuous Fourier spectrum, and the time-domain waveforms exhibit an aperiodic oscillation, which implies that the evolution of the cavity field is chaotic.

For all we know, the dynamic properties of the system can be achieved by observing the evolution of the nearby trajectories in phase space. Thus, the property of the cavity field is characterized by the evolution of an adjacent point $(I_{\rm a} + \delta I_{\rm a})$ of $I_{\rm a}$, and the Lyapunov exponent (LE) [69] of the cavity field is defined by the logarithmic slope of the perturbation $\delta I_{\rm a}$ ($\delta I_{\rm a} = |a + \delta a|^2 - I_{\rm a}$) versus the time t. In figure 3, we plot the evolution of the optical cavity field in phase space and $\ln \delta I_{\rm a}$ in temporal domain for three different values of the sphere-mirror separation d. As is clearly shown in figure 3(d), LE is negative, which demonstrates that the perturbation $\delta I_{\rm a}$ is decayed exponentially. Accordingly, the evolutional trajectory of cavity field in phase space with a tiny initial disturbance converges to a cycle called the classical limit cycle, which features the regular dynamic behavior of cavity field. The existence of attractor known as limit cycle in phase space which shows the periodicity of the trajectory, as shown in figure 3(a). Evidently, we can find from figure 3(e) that LE equals zero, which illustrates the emergence of period-doubling bifurcation, and the trajectory in phase space distinctly reveal the period-doubling oscillation of the cavity field as is shown in figure 3(b). Furthermore, we can clearly see from figure 3(f) that the LE is greater than zero, which implies the irregular dynamic behavior of cavity field and the evolution of the cavity field is very sensitive to the tiny changes of the initial conditions. The presence of a strange attractor reveals that the initial adjacent trajectory in the phase space eventually evolve into totally diverse states, which features the complexity and unpredictability [70], that is, the cavity field becomes chaotic and the trajectory of chaotic motion in phase space is uncertain, as shown in figure 3(c). Based on the above discussions, we discover a high dependence of the dynamic characteristics of the cavity field on the static Casimir effect. Our numerical calculation with experimentally attainable parameters indicates that as we reduce the sphere-mirror separation d, this means that when we increase the static Casimir effect, the dynamic behavior of the cavity field experiences the conversion from stabilized behavior to periodic-doubling oscillation and ultimately enters the chaotic motion, even if the driven optical fields are comparatively weak. The underlying physical mechanism can be understood as follows: the nonlinearity strength of the system plays a significant role in the generation of optical chaos. The regular-chaos conversion of the cavity field dynamic behavior originates from the nonlinear interactions between the cavity field and the mechanical resonator, and the nonlinear interactions can be substantively modified by the static Casimir effect. Thus, the generation of optical chaos can be controlled by the static Casimir effect.

To illustrate the remarkable influence of the applied static Casimir effect on the degree of optical chaos, we plot the dependence of the LE on the sphere-mirror separation d in figure 4. As can be seen from figure 4, the whole area can be divided into two regions: the blue region represents the order region and the white region indicates the chaotic region, separately. Obviously, the optomechanical system can be well-tuned to enter or exit the chaotic region by only harnessing the applied static Casimir effect without changing other parameters of the system, which provides us an effective and feasible way to manipulate the conversion from regular to chaotic motion, and has a vital significance in wireless Communication [71]. As figure 4 shown, in the regime of $d >$ 3.5 nm, the LE is negative, meaning the dynamic evolution of

cavity field is periodic. Interestingly, when $d = 3.5$ nm, the LE is zero, indicating the period-doubling oscillations of the cavity field. Furthermore, in the regime of d/nm $\in (2.3, 2.5)$, the positive LE characterizing the optical chaos. When the static Casimir effect does not exist ($d = \infty$), the motion of the system can be regarded as the superposition of two independent motions due to the intrinsic weak nonlinearity of optomechanical interaction. The linear terms in equation (2) play a dominant role in the evaluation of the system, hence the motion of the system is periodic. When the static Casimir effect exists, the nonlinear effect of the system can be enhanced, which leads to the frequency division (the first period-doubling bifurcation at $d = 3.5$ nm) of the system oscillation. LE is equal to zero only at the period-doubling bifurcation point [55]. When the static Casimir effect increases to a certain critical value ($d = 2.5$ nm), the nonlinear terms in equation (2) are strong enough to be comparable to that of the linear terms in equation (2). The competition between the linear and nonlinear terms in equation (2) becomes strong and the motion becomes very complex, thus forming chaos. When the static Casimir effect increases further ($d < 2.5$ nm), the linear terms in equation (2) become dominant again. Then the motion of the system becomes periodic. Figure 4 shows that the dynamic behavior of cavity field experiences the conversion from regular to chaos motion under the condition of properly adjusting the static Casimir effect, even if the driven optical fields are comparatively weak. To verify the numerical results of the cavity field evolution are reliable, we present the oscillation of the mechanical resonator with mechanical displacement x in the temporal domain for three different values of the sphere-mirror separation $d = \infty$, $d = 3.5$ nm, $d = 2.5$ nm, respectively, as shown in the inset in figure 4. Synchronization with the evolution of cavity field, the evolution of mechanical displacement x experiences the conversion from period (pink curve with $d = \infty$) to period doubling (green curve with $d = 3.5$ nm) and ultimately enters the chaos motion (blue curve with $d = 2.5$ nm) via properly adjusting the static Casimir effect. To more clearly show the chaotic motion, the evolutional trajectory of the mechanical resonator in phase space is also shown in the inset (located in the upper right corner of the figure 4) in figure 4. We can easy to see that the dynamic evolution of the mechanical displacement is chaotic and its motion evolves within a physical regime, which further verifies that the numerical results of the cavity field evolution are right and reliable. We see that the largest value of x is 25 pm, so that $x/d \lesssim 10^{-2}$ for $d = 2.5$ nm, and in this case the static Casimir potential energy can be expanded as power series $4\pi^3\hbar c R x^3/(720d^5) \sim 1 \times 10^{-22}$ J. Since this is already smaller that the other relevant energy scales in the Hamiltonian, this estimation suggests that taking into account only the x^3 corrections would be sufficient. Under this approximation, the optical chaos induced by the static Casimir effect comes from nonlinear terms $12V_{\rm sp}x^2/d^5$, $\hbar g(a_{\rm r}^2 + a_{\rm i}^2)$, $-gxa_{\rm i}$, $gxa_{\rm r}$ in equation (2). This investigation may provide a broad platform for the application of static Casimir effect in electro-optomechanics.

To study the evolution of the system more clearly from the interplay of the energy scales, the oscillation of the cavity field in the frequency domain and temporal domain for

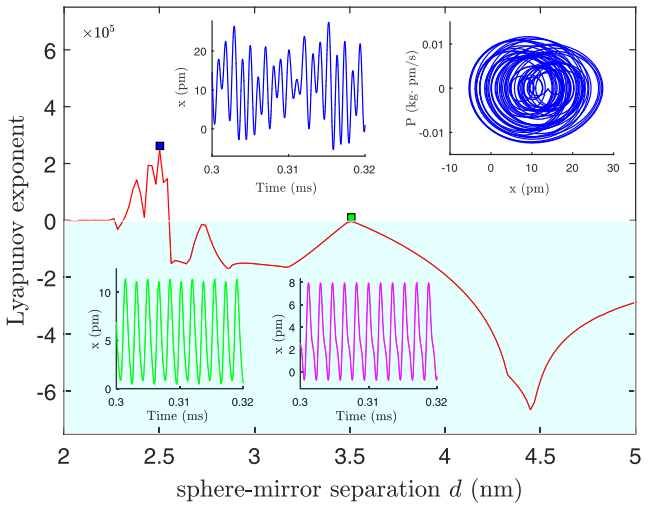


Figure 4. The LE varies with the sphere-mirror separation d within a fixed time interval $0.2 \rightarrow 0.5$ ms. The whole area is divided into two regions: blue area (order region) and white area (chaotic region). The insets: oscillation of the mechanical resonator in the temporal domain for three different values of the sphere-mirror separation d (pink, green, and blue curves denote $d = \infty$, $d = 3.5$ nm, and $d = 2.5$ nm, respectively), and the evolutional trajectory of the mechanical resonator in phase space for $d = 2.5$ nm (located in the upper right corner). The other parameters are the same as those in figure 2.

different values of the ratio $g_{\rm o}/\kappa$ with $g_{\rm o} = g\sqrt{\hbar/2m\omega_{\rm m}}$ are plotted in figure 5 We can clearly see from figures 5(b) and (f) that the period-doubling bifurcation oscillation occurs when the ratio $g_{\rm o}/\kappa = 8.0452 \times 10^{-5}$. With the decrease of the ratio $g_{\rm o}/\kappa$, the period-doubling bifurcation oscillation of the cavity field becomes periodic oscillation (shown in figures 5(a) and (e)), which means that the evolution of the cavity field is regular. Conversely, with the increase of the ratio $g_{\rm o}/\kappa$, the oscillation of cavity field is still period-doubling bifurcation oscillation, and the oscillation period is multiplied in turn (shown in figures 5(c) and (g)). As the ratio $g_{\rm o}/\kappa$ increases further, an aperiodic oscillation of the cavity field appears, which implies that the evolution of the cavity field is chaotic, such a result can be seen in figures 5(d) and (h). The underlying physical mechanism of these phenomena can be explained as follows: as one of an important system parameter, the ratio $g_{\rm o}/\kappa$ plays an important role in the process of energy exchange between the cavity field and the mechanical oscillator. The energy exchange between the cavity field and the mechanical oscillator is enhanced with the increase of the ratio $g_{\rm o}/\kappa$, that is, the increase of the ratio $g_{\rm o}/\kappa$ is beneficial to the enhancement of the nonlinear effect of the system. In contrast, the energy exchange between the cavity field and the mechanical oscillator is weakened with the decrease of the ratio $g_{\rm o}/\kappa$, which implies that the decrease of the ratio $g_{\rm o}/\kappa$ is detrimental to the enhancement of the nonlinear effect of the system. For the purpose of a more systematic understanding of the evolution of the system, the oscillation of the cavity field in the frequency domain and temporal domain for different values of the ratio $g_{\rm o}/\gamma_{\rm m}$ with $g_{\rm o} = g\sqrt{\hbar/2m\omega_{\rm m}}$ are also plotted in figure 6. We can see from figure 6 that the results are

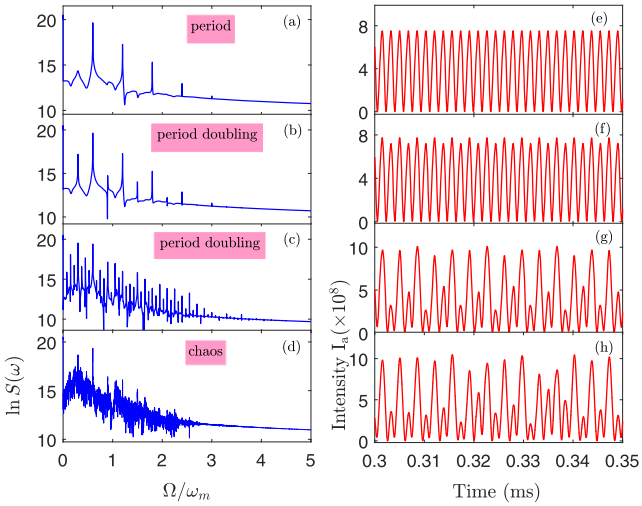


Figure 5. Oscillation of the optical cavity field is plotted in the frequency domain (a)–(d) as well as in the temporal domain (e)–(h), for four different values of the $g_o/\kappa = 7.3138 \times 10^{-5}$, $g_o/\kappa = 8.0452 \times 10^{-5}$, $g_o/\kappa = 1.2377 \times 10^{-4}$, $g_o/\kappa = 1.3409 \times 10^{-4}$ with $g_o = g\sqrt{\hbar/2m\omega_m}$, $d = 3.5$ nm, respectively. The other parameters are the same as those in figure 2.

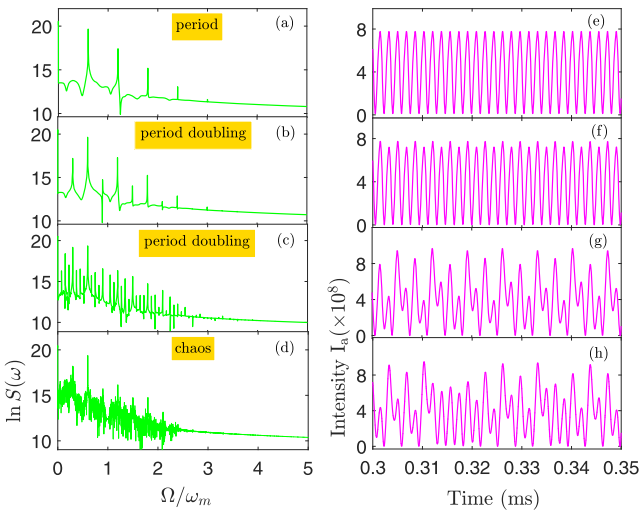


Figure 6. Oscillation of the optical cavity field is plotted in the frequency domain (a)–(d) as well as in the temporal domain (e)–(h), for four different values of the $g_o/\gamma_m = 0.1165$, $g_o/\gamma_m = 0.1227$, $g_o/\gamma_m = 0.1391$, $g_o/\gamma_m = 0.1411$ with $g_o = g\sqrt{\hbar/2m\omega_m}$, $d = 3.5$ nm, respectively. The other parameters are the same as those in figure 2.

similar to those in figure 5. It is shown that the dynamic behavior of the cavity field experiences the conversion from period to period-doubling bifurcation and finally enters chaos motion by increasing the ratio g_o/γ_m.

In the end, to understand the influence of the static Casimir effect on the optical threshold of chaos more systematically and comprehensively, the LE versus the control field power P_l for a fixed sphere-mirror separation $d = 2.5$ nm is plotted in figure 7. We can clearly see from figure 7 that the evolution of cavity field is characterized by chaotic motion when the

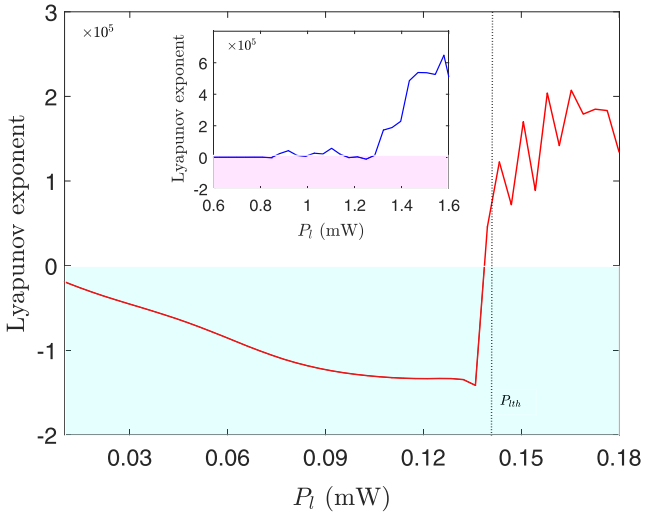


Figure 7. The LE varies with the control field power P_l for a fixed sphere-mirror separation $d = 2.5$ nm. The whole area is divided into two regions: blue area (order region) and white area (chaotic region). The inset: LE varies with the control field power P_l for a sphere-mirror separation $d = \infty$. The other parameters are the same as those in figure 2.

control field power $P_l > P_{lth}$. (P_{lth} indicates the optical threshold of chaos, which is defined by the control field power where the LE is zero). Similarly, in the inset in figure 7, we show the LE versus the control field power P_l for the sphere-mirror separation $d = \infty$ (static Casimir effect does not exist), which means that our system is reduced to a standard bare optomechanical system. Compared with the standard bare optomechanical system, we find that the optical threshold P_{lth} of chaos is reduced about one order due to the presence of static Casimir effect in our practical scheme. This provides an effective method to reduce the optical threshold P_{lth} of chaos. We note that in the current experiments, the accurate measurement of static Casimir effect for $d \sim 100$ nm has been realized [72]. Actually, the measurement of static Casimir effect for $d = 2.5$ nm is still facing challenges, however, even in the case of large d, strong static Casimir effect can be achieved by changing the optical properties, geometric structures, or the surface roughness of the interacting materials [73, 74]. For instance, for parallel graphene layers with very short separation ($d < 10$ nm), it was demonstrated that the Casimir force $\sim d^{-5}$. In the case that the parallel graphene layers with $d < 10$ nm has particular nanostructures, the Casimir force $\sim d^{-7}$ can be obtained [75]. This is sufficient to show that the required static Casimir effect with $d \sim 1$ nm for ideal metals can be realized for very larger separation with $d \sim 10$ nm or even 50 nm via properly changing the material properties. In addition, with the development of nanotechnology, a smaller separation can be expected in the near future.

4. Conclusion

In conclusion, we have investigated theoretically the generation of optical chaos in an optomechanical system driven by a strong control field and a weak probe field, and in which

the mechanical resonator interacts with the nearby gold-coated nanosphere via static Casimir effect. We find that such a system can induce the generation of optical chaos when the nonlinear strength is strong enough and reaches the threshold of chaotic. In particular, we reveal that the dynamic behavior of the cavity field experiences the conversion from period to periodic-doubling oscillation and ultimately enters the chaotic motion under the condition of properly adjusting the static Casimir effect, even if the driven optical fields are comparatively weak. Furthermore, our numerical calculation with experimentally attainable parameters also shows that the optical threshold of chaos can be reduced about one order compared with the standard bare optomechanical system due to the presence of the static Casimir effect. We believe that controlling chaos may have great significance for excavating the nonlinear characteristics of the optomechanical system. This investigation may open up an effective avenue for the study of static Casimir effect triggered chaos and find potential applications in chaos communication.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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